

Weak convolution continuity fails in every nontrivial Våge space

Candidate full negative answer to a question in arXiv:1106.5746

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Status: candidate full negative answer, likely valid. On a Våge space from the source, convolution is jointly continuous for the weak topology if and only if the generating set is empty. Hence every nontrivial example has strongly continuous but not jointly weakly continuous convolution.

1 Source question and setup

Alpay and Salomon prove in Theorem 3.6 that convolution is jointly continuous for the strong topology of a Våge space. Immediately afterward they ask whether it is continuous in the weak topology [1, p. 13].

We recall only the needed definitions. Let $A \subseteq \mathbb{N}$ and

$$\ell_A = \mathbb{N}_0^{(A)} = \bigoplus_{j \in A} \mathbb{N}_0 e_j$$

be the free commutative monoid. For a regular admissible superexponential weight $a = (a_\alpha)_{\alpha \in \ell_A}$, so that

$$a_0 = 1, \quad a_{e_j} > 1, \quad a_\alpha a_\beta \leq a_{\alpha+\beta},$$

the source defines

$$\mathcal{F}_a = \bigcap_{p \in \mathbb{N}} \ell_{a^p}^2, \quad \mathcal{F}'_a = \bigcup_{p \in \mathbb{N}} \ell_{a^{-p}}^2.$$

The convolution on $\mathcal{F}'_a$ is formal-series multiplication,

$$(fg)_\gamma = \sum_{\alpha+\beta=\gamma} f_\alpha g_\beta.$$

The weak topology in the question is $\sigma(\mathcal{F}'_a, \mathcal{F}_a)$, as is also clear from Proposition 3.7 and its proof in the source.

2 Proof intuition

Weak neighborhoods see only finitely many linear functionals. Consequently, a scalar bilinear form that is jointly continuous for two weak topologies must factor through finite-dimensional quotients; in particular, it has finite algebraic rank.

If convolution were jointly weakly continuous, pairing the product fg with any test vector $\varphi \in \mathcal{F}_a$ would therefore give a finite-rank bilinear form. Along any one generator $e \in A$, that form has the Hankel matrix $(\varphi_{(m+n)e})_{m,n \geq 0}$. We choose φ to decay so rapidly that it belongs to $\mathcal{F}_a$ but its Hankel matrix has infinite rank. This contradicts weak joint continuity.

Thus, $f_\lambda \rightarrow f$ in the strong topology of ℓ_{a-p}^2 . The opposite direction is clear. $\square$

Theorem 3.6. *Let $\mathcal{F}'_a$ be a Våge space. Then the convolution is a continuous function $\mathcal{F}'_a \times \mathcal{F}'_a \rightarrow \mathcal{F}'_a$ in the strong topology. Hence $(\mathcal{F}'_a, +, *)$ is a topological $\mathbb{C}$ -algebra.*

Proof. Assuming $((f_\lambda, g_\lambda))_{\lambda \in \Lambda}$ is a net which converges to (f, g) in the strong topology of $\mathcal{F}'_a \times \mathcal{F}'_a$, then in particular, $f_\lambda \rightarrow f$ and $g_\lambda \rightarrow g$ in the strong topology of $\mathcal{F}'_a$. According to Proposition 3.5, there exist $p, q \in \mathbb{N}$ such that $f_\lambda, f \in \ell_{a-q}^2$ and $g_\lambda, g \in \ell_{a-q}^2$ where $f_\lambda \rightarrow f$ in the strong topology of ℓ_{a-p}^2 and $g_\lambda \rightarrow g$ in the strong topology of ℓ_{a-q}^2 . We may assume that $p \geq q + d$. Since $\mathcal{F}'_a$ is a Våge space, $f g_\lambda = f * g_\lambda, f g = f * g \in \ell_{a-p}^2$, and $*$: $\ell_{a-q}^2 \times \ell_{a-p}^2 \rightarrow \ell_{a-p}^2$ is continuous. Since $(f_\lambda, g_\lambda) \rightarrow (f, g)$ in the strong topology of $\ell_{a-q}^2 \times \ell_{a-p}^2$, $f_\lambda g_\lambda \rightarrow f g$ in the strong topology of ℓ_{a-p}^2 . Again, using Proposition 3.5, we have that $f_\lambda g_\lambda \rightarrow f g$ in the strong topology of $\mathcal{F}'_a$. Thus the convolution is strongly continuous. $\square$

We do not know if the convolution is continuous in the weak topology. We end this section with the weak topology analogue of Proposition 3.5.

Figure 1: The strong-continuity theorem and the open weak-topology question on page 13 of the source.

3 Negative answer

Lemma 1. *Let (E, F) be a separating dual pair. Every scalar bilinear form $B : E \times E \rightarrow \mathbb{C}$ that is jointly continuous on $\sigma(E, F) \times \sigma(E, F)$ has finite algebraic rank.*

Proof. Joint continuity at $(0, 0)$ gives basic weak neighborhoods

$$U = \{x : |u_i(x)| < \varepsilon, 1 \leq i \leq r\}, \quad V = \{y : |v_j(y)| < \delta, 1 \leq j \leq s\}$$

such that $|B(x, y)| \leq 1$ on $U \times V$, where $u_i, v_j \in F$. If $x \in \bigcap_{i=1}^r \ker u_i$, then $tx \in U$ for every scalar t . For $y \in V$, boundedness of $tB(x, y)$ for every t forces $B(x, y) = 0$. Since V is absorbing, this holds for every $y \in E$. Thus

$$\bigcap_{i=1}^r \ker u_i \subseteq \{x : B(x, y) = 0 \text{ for all } y \in E\}.$$

The map $x \mapsto B(x, \cdot)$ therefore factors through a quotient of dimension at most r , and B has finite rank. $\square$

Theorem 1. *Let $\mathcal{F}'_a$ be a Våge space associated with a regular admissible superexponential weight on ℓ_A . The convolution map*

$$* : (\mathcal{F}'_a, \sigma(\mathcal{F}'_a, \mathcal{F}_a)) \times (\mathcal{F}'_a, \sigma(\mathcal{F}'_a, \mathcal{F}_a)) \longrightarrow (\mathcal{F}'_a, \sigma(\mathcal{F}'_a, \mathcal{F}_a))$$

is jointly continuous if and only if $A = \emptyset$.

Proof. If $A = \emptyset$, then $\ell_A = \{0\}$ and $\mathcal{F}'_a = \mathbb{C}$, so the assertion is immediate.

Suppose that $A \neq \emptyset$ and fix a generator $e = e_j$. Put $c = a_e > 1$. Superexponentiality gives, by induction,

$$a_{ne} \geq c^n \quad (n \geq 0). \quad (1)$$

Define a test vector $\varphi = (\varphi_\alpha)_{\alpha \in \ell_A}$ by

$$\varphi_0 = 1, \quad \varphi_{ne} = a_{ne}^{-n} \quad (n \geq 1), \quad \varphi_\alpha = 0 \quad \text{otherwise.}$$

For each fixed $p \in \mathbb{N}$,

$$\sum_{\alpha \in \ell_A} |\varphi_\alpha|^2 a_\alpha^p = 1 + \sum_{n \geq 1} a_{ne}^{p-2n} < \infty.$$

Indeed, only finitely many terms have $2n \leq p$, while (1) gives

$$a_{ne}^{p-2n} \leq c^{-n(2n-p)} \quad (2n > p).$$

Hence $\varphi \in \mathcal{F}_a$.

Assume for contradiction that convolution is jointly weakly continuous. The continuous functional

$$\Lambda_\varphi(h) = \sum_{\alpha \in \ell_A} h_\alpha \overline{\varphi_\alpha}$$

then makes

$$B_\varphi(f, g) = \Lambda_\varphi(fg)$$

a jointly weakly continuous scalar bilinear form. Lemma 1 says that B_φ has finite rank.

Let δ_α denote the coordinate vector at α and set

$$s_n = \varphi_{ne} \quad (n \geq 0).$$

On the linear span of $(\delta_{ne})_{n \geq 0}$, the matrix of B_φ is the Hankel matrix

$$B_\varphi(\delta_{me}, \delta_{ne}) = s_{m+n}. \quad (2)$$

This matrix has infinite rank. Otherwise, finitely many of its columns would be linearly dependent, so there would be constants $d_0, \dots, d_r$, not all zero and with $d_r \neq 0$, such that

$$\sum_{k=0}^r d_k s_{n+k} = 0 \quad (n \geq 0).$$

Thus the ordinary generating function $S(z) = \sum_{n \geq 0} s_n z^n$ would be rational. On the other hand,

$$s_n^{1/n} = a_{ne}^{-1} \leq c^{-n} \longrightarrow 0,$$

so S is entire. A rational entire function is a polynomial, which would make $s_n = 0$ eventually. This contradicts $s_n = a_{ne}^{-n} > 0$ for every $n \geq 1$. Hence (2) has infinite rank, contradicting Lemma 1. $\square$

4 Concrete one-generator witness

Take $A = \{1\}$ and $a_n = 2^n$. This is an exponential regular admissible weight and hence defines a Våge space by Theorem 3.3 of the source. The test vector above becomes $\varphi_n = 2^{-n^2}$ and lies in $\mathcal{F}_a$ because

$$\sum_{n \geq 0} |\varphi_n|^2 a_n^p = \sum_{n \geq 0} 2^{-2n^2 + pn} < \infty \quad (p \in \mathbb{N}).$$

Every leading Hankel minor is visibly nonzero. With $q = 1/4$,

$$2^{-(m+n)^2} = 2^{-m^2} q^{mn} 2^{-n^2},$$

and therefore the $(N+1) \times (N+1)$ determinant is a nonzero diagonal multiple of

$$\det(q^{mn})_{m,n=0}^N = \prod_{0 \leq j < k \leq N} (q^k - q^j) \neq 0.$$

The accompanying exact checker verifies this factorization through size seven; the displayed Vandermonde identity is the proof for all sizes.

5 Strength, scope, and novelty

The result is stronger than a single counterexample: it rules out joint weak continuity for every Våge space in the source whose free monoid has at least one generator, and identifies the sole trivial exception. It does not assert failure of separate weak continuity or of weak sequential continuity; these are strictly weaker properties and are not the question answered here.

Bounded searches on 11 August 2026 covered the local run indexes and the web, using the arXiv identifier, paper title, the exact quoted sentence, and the phrases “Våge space”, “convolution”, and “weak topology”. They found the source and adjacent work on other convolution topologies, but no later answer to this question or this finite-rank Hankel obstruction. Novelty confidence is moderate because the weak-bilinear finite-rank lemma is elementary and may be known independently.

References

- [1] D. Alpay and G. Salomon, *New Topological C-Algebras with Applications in Linear Systems Theory*, arXiv:1106.5746v2 (2012), <https://arxiv.org/abs/1106.5746>.